It is easy to observe that every identity function is one-to-one function as well as onto. A constant function is not onto unless the co-domain contains only one element. The following statements are some important simple results.

Let A and B be two sets with m and n elements.

(i) There is no one-to-one function from A to B if $m > n$.
(ii) If there is an one-to-one function from A to B, then $m \leq n$.
(iii) There is no onto function from A to B if $m < n$.
(iv) If there is an onto function from A to B, then $m \geq n$.
(v) There is a bijection from A to B, if and only if, $m = n$.
(vi) There is no bijection from A to B if and only if, $m \neq n$.



A function which is not onto is called an into function. That is, the range of the function is a proper subset of its co-domain. Let us see some illustrations.

(1) $X = \{1, 2, 3, 4\}, Y = \{a, b, c, d, e\}$and $f = \{(1, a), (2, c), (3, e), (4, b)\}$.
This function is one-to-one but not onto.

(2) $X = \{1, 2, 3, 4\}, Y = \{a, b\}$ and $f = \{(1, a), (2, a), (3, a), (4, a)\}$.
This function is not one-to-one; it is not onto.

(3) $X = \{1, 2, 3, 4\}, Y = \{a\}$ and $f = \{(1, a), (2, a), (3, a), (4, a)\}$.
This function is not one-to-one but it is onto. It seems that this function is same as the previous one. The co-domain of the function is very important when deciding whether the function is onto or not.

(4) $X = \{1, 2, 3, 4\}, Y = \{a, b, c, d, e\}$ and $f = \{(1, a), (2, c), (3, b), (4, b)\}$.
This function is neither one-to-one nor onto.

(5) $X = \{1, 2, 3, 4\}, Y = \{a, b, c, d\}$ and $f = \{(1, a), (2, c), (3, d), (4, b)\}$.
This function is both one-to-one and onto.

(6) $X = \{1, 2, 3, 4\}, Y = \{a, b, c, d, e\}$ and $f = \{(1, a), (2, c), (3, e)\}$.
This is not at all a function, only a relation.

(7) Let X be a finite set with k elements. Then, we have a bijection from X to $\{1, 2, \ldots k\}$.

Let us consider functions defined on some known sets through a formula rule.

Example 1.14 Check whether the following functions are one-to-one and onto.

(i) $f : \mathbb{N} \to \mathbb{N}$ defined by $f(n) = n + 2$.
(ii) $f : \mathbb{N} \cup \{-1, 0\} \to \mathbb{N}$ defined by $f(n) = n + 2$.

Solution:

(i) If $f(n) = f(m)$, then $n + 2 = m + 2$ and hence $m = n$. Thus f is one-to-one. As 1 has no pre-image, this function is not onto. (See Figure 1.30)
(ii) As above, this function is one-to-one. If m is in the co-domain, then $m - 2$ is in the domain and $f(m - 2) = (m - 2) + 2 = m$; thus m has a pre-image and hence this function is onto. (See Figure 1.31)